Student Fee Defaulter System

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Abstract:. In this paper we present a web based Student Fee Defaulter Management System that we put together to manage fee records, track outstanding fees, identify those that are behind in payment and at the same time we automated communication related to payment. We used Next.js and TypeScript for the interface and Java Spring Boot, and Python Flask for services which all are tied in with a shared Supabase PostgreSQL database. Also students can pay fees via Razorpay or they can put in bank payment receipts for off line verification. We used Gemini 2.5 Flash to get out the transaction info from the uploaded receipts which in turn are looked at by admins. For automation we used n8n and Brevo and also we sync up payment details with Google Sheets. We have put together a single platform which includes fee tracking, payment processing, receipt verification, and notification management which in turn we have made a central point for students' fee issues.

Index Terms— Student Fee Management, Fee Outster System, Payment Tracking, Automatic Notifications, Receipt Valuation, OCR, Online Payment, Workflow Automation.

I. INTRODUCTION

Fee management is a primary function of what educational institutions do which includes maintaining student records, assigned fees, collection and of past due amounts. What is done present is a separate record keeping which in turn makes fee tracking a complex issue and increases admin effort. We have developed the Student Fee Defaulter Management System which brings all of that into one web based platform. The system we present allows admins to manage student records, set fees, track payments, to identify past due or only partially paid accounts and to also send out fee related notifications. We have developed a centralized supabase PostgreSQL database to store student data, fee structure, and receipt details. Additionally, the online payment method via Razorpay has been incorporated into the system, while the offline payment method offers the students an option to upload their

net banking receipts. Besides this, we have employed Gemini 2.5 Flash OCR to extract the UTR, amount paid, and the date of the transaction from the uploaded receipt, which is then subjected to manual verification by the admin.

The system makes use of n8n and Brevo to send out various SMS and Emails pertaining to OTP verification, payment confirmation, reminder for pending fees, and challan-status update, among others. It is also possible to log the payments made into a Google sheet through automation.

The application has been built using Next.js and Typescript alongside Java Spring Boot and Python Flask backend services. The application has various security features such as OTP and password protection, restriction on the number of attempts to log in, and temporary account locking. In essence, this platform offers a centralized solution for managing and tracking fee payments, processing payments, verifying receipts, and handling defaulters.

II. PROBLEM STATEMENT

The management of the students' fee records, their payments, and unpaid dues with their due dates is challenging when managing the fee records of the students, their payments, and the unpaid amounts and their due dates as separate procedures. The former requires the monitoring of an increasing number of students with their payments and unpaid dues. It is challenging to manage and monitor them.

Their payment verification is an additional procedure where the students pay their fees online and offline, and the institution automatically or manually verifies their receipts.

Furthermore, their communication is an additional procedure as the institution seeks to remind them of their unpaid, partially paid and confirmations of their online payments and receipt submissions.

Hence, it is essential to have an integrated system to manage their fee records, monitor their unpaid, partially paid fees, manage their online and offline payments, verify their receipts, and automate their communication. The proposed Student Fee Defaulter Management System manages to manage all the procedures mentioned above via an integrated web application.

III. LITERATURE REVIEW

The proposed system manages digital fee management by organizing the students' fee records and managing their payments conveniently.

Monitoring their unpaid and partially paid fees is essential to manage the former's collection and is stored in a database accessible via the aforementioned application.

The proposed Student Fee Defaulter Management System enables students to make their payments online and stores the latter's records in the aforementioned database via Razorpay to collect the former digitally.

The students also make their payments offline and submit their bank receipt or challan from which Gemini 2.5 Flash OCR extracts the UTR, amount, and date of the payment for the institution's manual verification to update the students' fee records.

Automation of communication eliminates the need to remind the students of their unpaid, partially paid fees, and confirmation of their online payments and receipt submissions.

It uses n8n and Brevo to send SMS and Emails to the students for their OTP verification, overdue and partial payment reminder, confirmation of online payments, and receipt submissions. The aforementioned web application schedules the aforementioned tasks to remind the students of their unpaid, partially paid fees. The Next.js, TypeScript, Java Spring Boot, Python Flask, and Supabase PostgreSQL, payment, OCR, and email verification services comprise the proposed solution, which utilizes robust security features to protect students' data, including OTP, strong password hashing, limited login attempts, and temporary account lockout.

V. SYSTEM ARCHITECTURE

The proposed Student Fee Defaulter Management System has been designed using the layered architecture that has Client/User, Application and Automation, Central Database, AI and Payment Gateway, and Notification and Logging Pipeline layer. The layer division helps the individual components of the system to perform the specified functions and connection through application services and central database offers the combined working of the system.

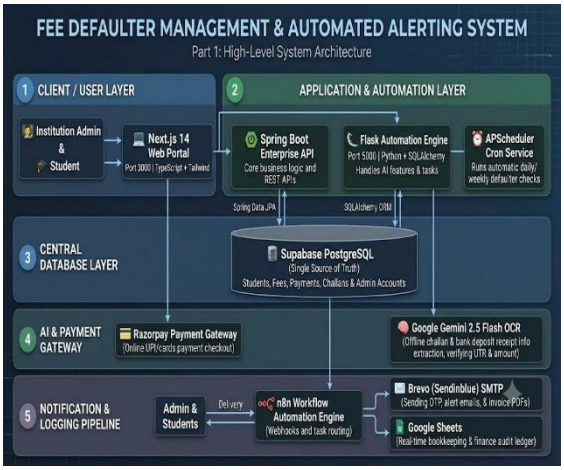


Fig1.System Architecture of the Proposed Student Fee Defaulter Management System

The Client/User Layer depicts the users of the system such as institution administrator and students. The users can access the application through Next.js 14 web portal that has been developed using the TypeScript and Tailwind CSS. The portal offers the interface to access the features offered for the different categories of the users.

The Application and Automation Layer constitutes the main processing layer of application architecture. It incorporates Spring Boot Enterprise API, Flask Automation Engine, and APScheduler Cron Service. While the Spring Boot service offers the fundamental REST APIs and application level operations, the Flask service offers the automation and AI functionalities. The APScheduler is utilized to offer the pre-defined scheduled task such as recurring fee and default checking.

The Central Database Layer offers the common data storage of the application. The system utilizes the Supabase PostgreSQL database to store the essential records such as student data, fee

information, payment records, challan details, and administrator accounts. The database acts as a common storage that can be accessed by the application services.

The AI and Payment Gateway Layer accommodates the external specialized services that are required by the application. The Razorpay Payment Gateway has been integrated to offers the online fee payment modes such as UPI and cards. The Google Gemini 2.5 Flash OCR has been applied to process the uploaded offline challans, bank deposits, and extract the relevant payment details. The two external services offer the extended functionalities of fee payment and AI-based document analysis to the system.

The Notification and Logging Pipeline offers the automated communication and additional logging of payment records of the system. The n8n Workflow Automation has been employed to link the events with appropriate automated tasks. The Brevo SMTP has been utilized to offer email services such as OTP, fee alerts, and other notifications. The Google Sheets has been applied to maintain the additional logging sheet as the bookkeeping or audit log.

As depicted in Fig 1, the five layers represent the system architecture of the proposed solution. The architecture consists of the components that have been divided based on the individual functions while establishing the connections that enable the combined working of the system.

VI. WORKING OF THE PROPOSED SYSTEM

The proposed Student Fee Defaulter Management System offers separate portals to administrator and students while providing online and offline modes for the fee payment. The working process of the system involves system initialization and fee setup, fee monitoring and defaulting, student authentication, online and offline fee submission, receipt verification, and notifications. The overall working process of the system has been depicted in Fig. 2.

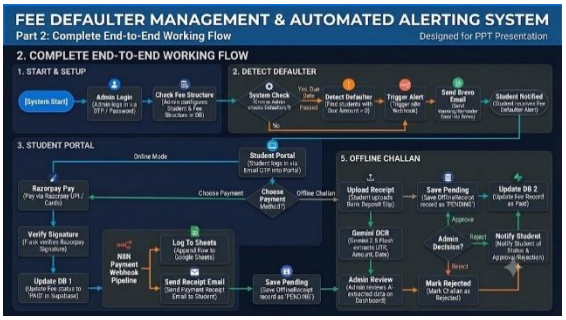


Fig 2.End to End Working flow

A. System Initialization and Fee Setup

The working process begins with the administrator authentication. The administrator is allowed to access the fee-related features of the system on providing the required credential. The administrator can set the required fee structure and provide the fee details to the students. The stored fee details are later utilized to track the fee status of the students.

B. Fee Monitoring and Defaulter Identification

The system maintains the track of fee status of the enrolled students. An automated scheduled checking process tracks the due fee details and recognizes the defaulters on checking if still due beyond the due date. The identified defaulters are added to defaulter list to notify regarding the due fee.

C. Student Authentication and Portal Access

The students are required to authenticate to access the Student Portal by providing the registered roll number and email address.

The student authentication process involves OTP-based verification to offer restricted access to the student fee details. The successful authentication redirects the student to student dashboard.

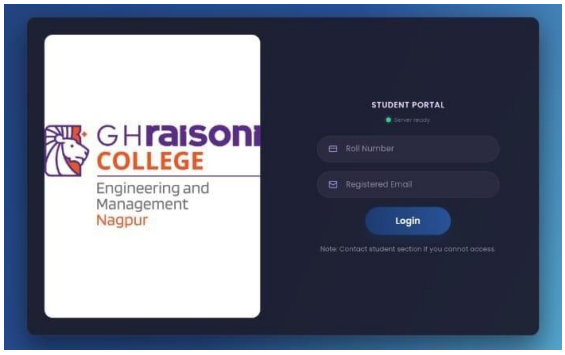


Fig 3. Student Portal Login Interface

The dashboard offers the relevant fee-related information such as the total course fee, amount paid, late fine, and the remaining amount that needs to be paid.

D. Online Fee Payment

The student can select the online mode to pay the fee. The portal shows the amount that needs to be paid, and the student can choose to proceed with the online payment option. After the payment is completed, the payment details are verified before the fee is marked as paid.

On successful verification, the fee is updated as paid, and an email confirmation can be sent to the student. The payment process can also be recorded in the logging sheet using the automated logging process.

E. Offline Challan Submission and Verification

The system also allows students to pay the fee through an offline banking method. The student can submit the bank challan or receipt of the fee payment through the Student Portal.

F. Notification and Record Maintenance

The system will implement a workflow automation tool that will handle all communication regarding authentication, fee reminder, successful payments, and challan status. In addition to these, the data related to payments are also stored in the main database, and successful payments can also be stored using the Google sheets workflow. It will help to organize and store the student's fee data and payments methodically and orderly throughout the course of the fee payment process.

Therefore, the working system will provide the students with the options to pay the fees in two ways which are through online using the payment gateway and through the challan and administrative verification process that will be related to the student's fee record and payment database.

VII. RESULTS AND DISCUSSION

The designed Student Fee Defaulter Management System was supposed to solve the problem of student fee management, monitoring of payments, detection of defaulters, and associated operation administration in the web-based desktop application.

The application’s Administrator Dashboard demonstrated the general information about the student fee, the list of fully paid, partially paid, and default student accounts, late fees, and several actions to quickly resolve the issue of student defaulters.

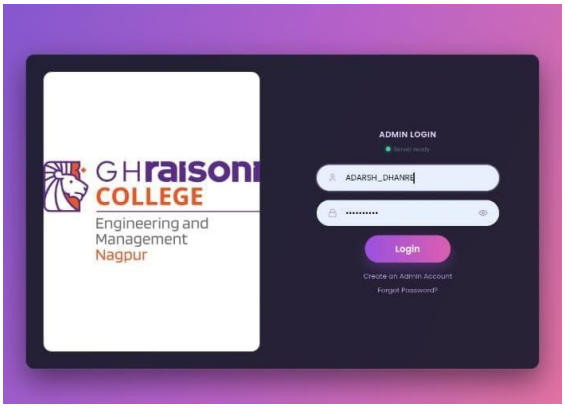


Fig 4.Administrator Dashboard

The Defaulter interface of the application displays the list of students who have not fully paid the assigned fee. The interface includes columns with the amount of the assigned fee, the amount paid by the student, the remaining balance, and information about the due date. Using this information, the administrator of the application can monitor the current student defaulters and remind them about the necessity to pay the remaining amount of the fee.

The Fee interface of the application allows the administrator to monitor the student fees and their current balances. The list of student fees includes the amount of the assigned fee, the amount paid by the student, the remaining balance, and late fees, if any. Therefore, the designed application resolves the problem of student fee management and defaulters by allowing the administrator to track the students' payments and remind them about the due date of payments. From the provided information, it could be concluded that the described application features effectively solve the identified problem and provide the necessary convenience in managing the student fee and defaulters.

VII. FUTURE SCOPE

The proposed Student Fee Defaulter Management System can be extended for larger institutional requirements. Future scope of this project can involve development of a dedicated mobile application so that students and administrators can conveniently access the fee management services using mobile devices. We can add more payment gateways and options to the system so that it provides more flexibility to the users.

Receipt verification module can be improved to support multiple receipt formats and to provide better result for unclear receipts. We can also add more graphical analysis and reports for the administrators to keep track of their payments, defaulters list, and other relevant data of the students.

The future version of the proposed solution can be integrated with existing system or software of the college or university to avoid extra data entry. We can also add new notification methods including SMS and other communication medium supported by the institute. These future enhancements will allow the proposed solution to deploy in larger scale for automation of student fee processing and tracking in advanced manner.

VIII. CONCLUSION

The Student Fee Defaulter Management System offers an integrated approach to managing student fee-related activities. Within the system, the Student Portal and Admin Panel are connected, enabling students to track their fee status, review payments, make online payments, and

upload offline receipts. In addition, the administrator can view student data, assign and track fee amounts, and identify students with outstanding balances or partial payments.

The integration of the Razorpay, Gemini 2.5 Flash, n8n, Brevo, and Google Sheets tools add additional functionality to the Student Fee Defaulter Management System, as they enable extra online payment options, extract bank receipt data to verify payments, and automate routine operations. For instance, the system can send OTP codes to students to confirm payments, send payment confirmation emails and text messages, and send reminders to delinquent payers. Overall, the tools mentioned above have been integrated into the system, enabling end-to-end payment processing and communication functions.

Overall, the created prototype showcases an efficient approach to centralizing student fee-related activities. Additionally, the system includes necessary security and safety features, such as OTP authentication, password protection, limited login attempts, and account lockout functions. Moreover, it offers an organized approach to student fee management, as it provides a set of necessary functions and can be expanded to include additional tools and features.

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